Identification Chart for the METAMORPHIC ROCK COLLECTION

Ward's Natural Science 45 E 3120 · Format adapted from Bob Leighty's Identification Chart for Common Metamorphic Rocks

TEXTURE	GRAIN SIZE	ROCK NAME	DESCRIPTION	INTERPRETATIONS		ADDITIONAL INFORMATION (setting & parent rock)
				Common Protoliths (original or "parent" rock)	Metamorphic Grade / Type	
Foliated (layered)	very fine	**RED SLATE**	dull, smooth, rings when struck; layers break into thin flat plates (slaty cleavage); **red-brown** from hematite	red shale, red clay	low pressure (P) & low temperature (T) metamorphism (regional)	Shale deposited in a basin is buried and squeezed at a convergent margin (about 200 to 300 °C). Red color: hematite (Fe^{3+}) in an oxidized sediment.
	very fine	**GRAY SLATE**	same slaty cleavage; **gray to black** from organic carbon in the original mud	black mud, shale with organic matter	low P & low T metamorphism (regional)	Same setting as red slate. Carbon from buried organic matter, not oxidized, gives the dark gray to black color.
	medium to coarse	**SCHIST**	layers of flaky, glistening **mica** or green chlorite (schistosity); no obvious extra minerals	mudstone, shale	low to moderate P & T metamorphism (regional)	Mudstone heated further in a mountain belt (about 350 to 500 °C); mica grows large enough to see. Chlorite marks lower grade, muscovite and biotite higher.
	medium to coarse	**GARNETIFEROUS MICA SCHIST**	mica schist with numerous **garnet** crystals (index mineral)	mudstone, shale	moderate P & T metamorphism (regional)	Garnet grows at moderate temperature (about 450 to 550 °C and up) in the zone of regional metamorphism in a collision belt. An index mineral for grade.
	medium to coarse	**HORNBLENDE SCHIST**	dark green to black; aligned **hornblende** (amphibole) crystals; minerals completely reformed and rearranged	basalt, gabbro (dark igneous rock)	moderate to high P & T metamorphism (regional); amphibolite facies	Basalt or gabbro (ocean crust) caught in a mountain belt; recrystallizes at about 500 to 700 °C into amphibole plus feldspar (amphibolite facies).
	coarse	**BIOTITE GNEISS**	coarse light and dark **compositional bands** (gneissic banding); dark **biotite** streaks; crushed and squeezed	granite (plutonic rock)	high P & high T metamorphism (regional)	Granite buried deep in a collision zone (about 600 °C or more); minerals segregate into bands. Gneiss from granite is an orthogneiss. Next step up: migmatite.
	coarse	**GRANITOID GNEISS**	light, granite-like quartz and feldspar with weaker **banding**	granite (plutonic rock)	high P & high T metamorphism (regional)	Same setting as biotite gneiss. Less biotite and weaker banding mean it still resembles its granite parent.
Non-Foliated (massive or no layering)	fine to coarse	**QUARTZITE**	interlocking quartz breaks across grains; **very hard (H = 7)**, scratches glass; will not react with HCl; original sand grains hard to distinguish	quartz sandstone (quartz arenite)	variable metamorphic grade; low P & high T (contact) or regional (no foliation: no platy minerals)	Quartz sandstone recrystallized; quartz grains grow together so the rock breaks through grains, not around them. Forms near intrusions or in mountain belts.
	medium to coarse	**WHITE MARBLE**	white, sugary interlocking **calcite**; easily scratched with a steel nail (H = 3); **fizzes in dilute HCl**	limestone	variable metamorphic grade; contact or regional	Limestone recrystallizes: calcite grains grow and interlock, destroying fossils and bedding. Forms around intrusions (contact aureoles) or in mountain belts.
	medium to coarse	**PINK MARBLE**	pink from impurities; fizzes if calcitic; if **dolomitic**, powder the surface before it reacts	limestone, dolostone	variable metamorphic grade; contact or regional	Same origin as white marble. Pink color from iron or manganese impurities. If the parent was dolostone, the rock is dolomitic and reacts weakly to acid.
	fine	**SERPENTINITE**	dark green, **waxy or greasy** luster; harder than a fingernail, softer than glass; no HCl reaction	peridotite (ultramafic rock)	low metamorphic grade; hydration of olivine and pyroxene to serpentine	Mantle peridotite reacts with water at low temperature (below about 500 °C) at mid-ocean ridges and subduction zones: olivine becomes serpentine. Found in ophiolites.
	fine	**SOAPSTONE**	soft (H = 1), **slippery, soapy feel**; mostly talc; scratched by a fingernail; may be weakly foliated	ultramafic rock altered by Mg-rich fluids	low to moderate grade; low P & T	Mg-rich ultramafic rock altered by hot, CO_2-bearing fluids; talc forms (about 300 to 500 °C). Soft and heat resistant, so used for carving and stoves.

HIGH ↓ metamorphic grade ↑ LOW

Tests: foliation (rotate under a light) · acid (one drop dilute HCl, goggles on) · hardness: fingernail 2.5, copper coin about 3.5, steel nail and glass about 5.5, quartz 7.

Grade sequence: slate, phyllite, schist, gneiss with rising T and P. Phyllite (fine, satiny sheen, wavy surfaces) is not in this collection. Name schists by prominent minerals (garnet, hornblende, etc.).